

Analysis First-Year Exam, August 2025, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. This problem has two parts.

- (i) Let $f : [0, 1] \rightarrow \mathbb{R}$ be Riemann integrable. Must the function F defined on $[0, 1]$ by $F(x) = \int_0^x f(t) dt$ be continuous? Explain.
- (ii) Let $g : [0, 1] \rightarrow \mathbb{R}$ be Lebesgue integrable. Must the function G defined on $[0, 1]$ by $G(x) = \int_{[0, x]} g d\lambda$ be continuous (λ : Lebesgue measure)? Explain.

2. Equip the set $C[0, 1]$ (comprising all continuous real-valued functions on the interval $[0, 1]$) with the sup metric and let

$$B = \{f \in C[0, 1] : \sup |f| \leq 1\}$$

be its closed ball of radius one about the zero function. Is B complete? Is B compact? Explain.

3. Assume that the continuous function $f : [0, 1] \rightarrow \mathbb{R}$ satisfies

$$\int_0^1 f(t) \cos(nt) dt = 0$$

whenever n is a non-negative integer. Does it follow that f is the zero function? Proof or counterexample required.

4. Let $(f_n)_{n=0}^\infty$ be a sequence of real-valued measurable functions on Ω . Prove that if $(f_n)_{n=0}^\infty$ converges pointwise to $f : \Omega \rightarrow \mathbb{R}$, then f is measurable.

5. The bounded function $f : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ is continuous in each variable separately: thus, $f(x, y)$ is continuous in x for each fixed y and continuous in y for each fixed x . Prove the continuity of F defined by

$$F(x) = \int_0^1 f(x, y) dy.$$